

OGC2026 Technical Report

Abstract. The challenge asks for a schedule and a physical layout at the same time: every block needs a bay, an entry time, and a pose that a vertically descending crane can actually reach. Our starting observation is that the objective $w_1Z_1 + w_2Z_2 + w_3Z_3$ reads only the pair (bay, entry time)—coordinates and orientation appear nowhere in it. Geometry is therefore a feasibility constraint that decides how much scheduling freedom exists, not a term to be optimised directly. The two training sets do not even share a shape — over half the preliminary instances carry no tardiness at all, none of the final forty do — so we avoid density thresholds and instance classification entirely: one portfolio of six operators competes for the budget on measured payoff per second, and every hot loop lives in one of two C++ extensions. Our main contributions are an exact contact upper bound that skips 40.8% of the candidate cells while provably returning the identical solution, a $10.5\times$ faster conflict-graph build verified by an unchanged edge count, and a position-independent geometric predicate that removed a floating-point inconsistency from the packing test. The most instructive result is negative: that same verified $1.85\times$ speedup made one instance 7.7% worse, because the operator schedule runs on a wall clock and a faster search therefore takes a different path rather than a longer one.

1 Introduction

1.1 Challenge Problem & Main Difficulties

Assigning hull blocks to bays over time is the spatial scheduling problem of shipbuilding. Three properties of the version posed here shaped everything we built.

The objective and the constraints read different variables. Tardiness, workload imbalance and preference are all functions of which bay a block occupies and when it enters; x , y and orientation appear in none of them. Geometry enters only through feasibility. A layout that cannot accommodate a block pushes its entry later, and that is the only channel through which packing quality becomes objective value. Treating tightness as an objective term is a modelling error we made early. Weighting contact more heavily does not buy a better schedule, it buys a denser layout whose scheduling consequences are unmeasured.

The crane makes the packing three-dimensional. A block is lowered vertically, so a placement is legal only if every layer of the descending block clears everything present along the whole descent. Two neighbours occupying the same floor cells but standing one and two layers high ration space completely differently. The scarce resource is height, not occupancy, and that is why the descent test goes layer by layer.

The two rounds do not have the same shape. At a uniform 60s the preliminary training set is bimodal: Z_1 is exactly zero on twenty-one of its forty instances and strictly positive on the rest (Table 1). The final-round set is not bimodal at all. All forty of its instances are late, and the share of the objective carried by tardiness spreads from 2.5% to 97.9% with no gap wider than fifteen points anywhere in it (Table 2). Where Z_1 vanishes the entry times are free and the problem is an assignment dominated by preference; where the yard is saturated they are the whole objective. Neither shape admits a threshold — in the first the classes are separate but the boundary is unobservable, in the second there are no classes — so we built a scheduler that measures which operator is paying rather than one that classifies the instance.

1.2 Contributions

- A *gate-free operator portfolio*: six operators compete for the budget on realised objective gain per second, with no density threshold anywhere; slice sizing is a separate question answered by completion.
- An *exact contact upper bound* skipping 40.8% of candidate cells while provably returning the identical solution, verified per cell rather than by comparing run outcomes.
- A 10.5× faster *conflict-graph build*, accepted on an unchanged edge count rather than a stopwatch, over a shared memo made safe by compare-and-exchange.
- *Offset-relative geometry*: the packing predicate became position-independent, fixing a disagreement we had misdiagnosed as hashing.
- A negative result more useful than any of these: an exact, verified 1.85× speedup that made one instance 7.7% worse (Section 2.3).

2 Algorithm Description

2.1 Overall Algorithm Framework

Four stages, and no preprocessing worth the name. *Initial solution*: each of four independent worker processes seeds a pool with a cheap always-feasible sequential schedule — the one guarantee that the algorithm never returns nothing, and the reason every other fallback path could be deleted. *Improvement*: each worker then runs an operator loop until its budget is exhausted, differing from the others in random seed and in its starting offset into six diversification axes, so the four explore different trajectories over the same operators. *Selection*: the best full objective across workers is taken, and the entry point reserves $\min(0.2T, 40\text{ s})$ of the limit for a final preference-repair pass over that winner. *Validation*: feasibility is enforced inside the operators and re-checked outside them — one acceptance function re-runs the provided checker on anything that beats the incumbent and returns $+\infty$ if it fails. An operator may be wrong about feasibility; a wrong answer cannot be accepted.

There are six operators. A contact-beam construction. An evolutionary re-growth with crossover and path-relinking over (bay, dispatch-order) anchors, decoded by the same beam. A workload-balance repair, a preference repair, a CP-SAT bay reassignment, and the exact bay repacking of Section 3. Each receives one probe to establish a rate, and the budget then goes to whichever has the highest realised objective gain per second. Fifteen percent is held back for exploration so a slow starter can recover. The arrangement is an adaptive large neighbourhood search [3], with one difference. Its adaptive weights score realised objective gain per second, where the usual form counts accepted moves.

Selection and sizing are different questions. Slice size is deliberately not driven by the selection signal, because sizing on payoff spirals downwards. A construction that returns a good solution but merely fails to beat the incumbent scores zero gain. It shrinks, and then it can no longer finish at all. We measured fourteen calls returning four solutions while the objective moved from 48,840 to 56,841. Size is a completion question instead: an operator that returned nothing was starved and is given more, one that returned anything fitted, and a repair pass is given what it used.

2.2 Key Ideas that Succeeded

Five changes moved the objective. They are given in the order they act, from the construction that produces a solution to the operator that repacks one.

Contact beam, and why its contact measure is flat. The construction is a filtered beam search over dispatch order [2], ranking survivors on the exact objective rather than a priority index. Each state places the next block at the position minimising a weighted sum of negative contact, a positional sweep term and a preference penalty, and survivors are completed by a rollout. Contact plays the role touching perimeter plays in two-dimensional packing, extended to a descent the crane must clear.

Contact admits two measures. A per-layer one describes the geometry better — a block beside a shorter neighbour earns nothing for the layers facing open air — and it is implemented, but the submitted configuration uses the flat measure because the bound below is derived against a flat occupancy prefix sum, which does not dominate per-layer occupancy. The two cannot both be on, and the bound is worth $1.09\text{--}1.85\times$. This is a constraint we accepted rather than a property we wanted; with a per-layer bound the choice would likely go the other way.

An exact contact upper bound. Profiling showed 86% of scored cells were evaluated after the best cell of that scan had already been found. Bounding the positional term fired zero times in 112.9 million cells. The score is contact-dominated by more than an order of magnitude, with ranges of about 2.3 against about 60, so a bound that knows only the sweep coordinate is never competitive.

Bounding the contact itself inverts this. Contact counts the four neighbours of each occupied footprint cell. A neighbour outside the bay scores one, a neighbour inside the same footprint scores nothing, and an occupied neighbour scores its weight. Every cell that can contribute therefore lies inside the footprint bounding box dilated by one, and counts at most once per direction, giving

$$\text{ct}(i_x, i_y) \leq W(i_x, i_y) + 4w_{\max} |\mathcal{O} \cap B^{+1}(i_x, i_y)|, \quad (1)$$

where W counts cell–direction pairs pointing out of the bay, $\mathcal{O}$ is the occupied set and B^{+1} the dilated box. An occupancy prefix sum, built once per (bay, time window) at the same asymptotic cost as the occupancy map it accompanies, answers the rectangle in four reads, and a cell whose best possible score cannot beat the incumbent is skipped without any geometric test. The bound is applied only where it is proved, which is what excludes the per-layer variant.

A cliff at the bottom of the beam, and the axis that came out of it. The beam checked its deadline at each level and returned nothing when it passed, so every worker fell back to the sequential floor. The failure was not gradual. On a 300-block instance the answer was 4.02×10^9 at 110 s against 9.69×10^7 at 130 s, a factor of 42, and 143 on a 250-block one, at a threshold that moves with instance size while the hidden limits are undisclosed. The repair needed no new machinery, since the beam already completes a partial state by rollout to rank survivors: on expiry it completes the incumbent rather than discarding it.

That exposes a knob rather than a patch: the fraction of its slice the beam claims before handing the remainder to the rollout. Sweeping it down improves large instances monotonically (−24.3% to −11.9% at one tenth) and harms small ones monotonically, since the beam finishes anyway there and a low aim only narrows it (+25.09% at 150 blocks). We did not resolve that by testing instance size. The workers are already a portfolio whose minimum is the answer, so half run the low aim and half the high one, each group carrying the instances the other loses, and nothing is gated. Over 37 paired final instances this is 31 better against 6 worse, median −3.44%, sign test $p < 0.0001$, splitting 17–2 on the large group and 14–4 on the small. Our prediction that the small group would pay was wrong: true of a constant, false of an axis. A portfolio does not need every member to be good, it needs them to fail on different instances.

Replicating the two small instances that lost by more than 10% shrank both to +3.2%. A single paired draw at 150 blocks is worth its sign, not its magnitude.

A conflict-graph build that watches its own clock. The repacking operator — which lifts every block out of one bay and re-seats the whole set exactly, and is described in Section 3 — begins with an $O(n_{\text{col}}^2)$ conflict graph that cannot be interrupted. Two columns conflict only while both are present, so sorting by entry time and breaking the inner loop once the later entry passes the earlier exit skips 83% of pairs without visiting them, and the fields the filter reads moved into flat arrays. Neither change can alter which pairs conflict. The edge count therefore had to come out identical, and that was the acceptance test, not the

clock. The same configuration now builds in 10.7–14.2 s against about 90 s before. The build also projects its own completion every 256 rows and steps down to a cheaper configuration it can finish rather than losing the call.

The rows are independent, so the loop is finally parallel. Where the results go is not independent. Each thread keeps its own edge list, and the memo on the conflict predicate is shared. Sharing it naively is not merely racy but silently wrong, because the table is open-addressed and two threads inserting different keys can both claim one empty slot, the loser then writing its verdict into the winner’s entry. A slot is therefore claimed by compare-and-exchange, with relaxed ordering, since the value is a pure function of the key. A private memo per thread gave almost all the parallel gain back (1.10×), reuse being where this loop’s speed lives, while sharing it correctly reaches 1.43× on four cores at an edge count of 133,863,098 in every arm.

An identical edge count is also what made the memo usable: its first version produced 36 extra edges, and dumping the disagreements showed the predicate itself was position-dependent through floating-point rounding, not the key aliasing we assumed. Comparing origin outlines plus an offset makes the verdict a function of relative position alone, fixing the memo by fixing the predicate.

Pricing a seat, not a block. The operator hands the solver a list of candidate placements, a *column* being one block at one orientation, position and entry time, and asks for the maximum-weight conflict-free selection. The weight was attached to the *block*. Since the objective reads only (bay, entry time) and one call concerns one bay, the entry variant is exactly the granularity at which a block’s alternatives differ: an on-time column and a ten-days-late column into a preferred bay carried the same number, so the solver took whichever seated more easily.

Every attempt to express the trade from outside failed for the same reason. Discounting a block’s weight by the cost of its late seat undervalues it when it takes the on-time one, and at the break-even delay, the only principled place to put the offer, the discount equals the entire gain: the weights of precisely the high-regret blocks collapse to the floor. Across the final instances, discounting is indistinguishable from never offering a late window at all: 9 better against 21 worse, median +0.00%.

Attaching the weight to the entry variant removes the workaround instead of tuning it. Each offered window is priced by the only term an entry time can move,

$$p(w) = g - w_1(\max(0, e_w - d) - \max(0, e_0 - d)), \quad (2)$$

with g the gain of the move, e_w the exit under window w , e_0 the current exit and d the due date. The expression is exact in both directions. A window *earlier* than the block’s current entry therefore prices above the base, and the solver can be paid for pulling a block forward. A per-block weight cannot represent that at all. On 39 final instances the priced form is 20 better against 9 worse (sign test $p \approx 0.03$, mean -0.74%), and it does no harm where no late window can open.

2.3 Key Ideas that Failed

A wall-clock schedule turns a speedup into a different answer. This is the result we would most like to pass on. Having verified that the contact upper bound is exact, that it returns bit-identical placements at fixed work, and that it is $1.85\times$, $1.66\times$ and $1.09\times$ faster on three instances of very different density, we enabled it and one instance got 7.7% worse.

The bound is not at fault, and neither is anything else in the search. The operator loop decides what to run next, and for how long, from elapsed seconds, so a faster scan fits more calls into the same budget and shifts the diversification axis rotation, the solution pool, and the repacking operator's call counter and hence its seed. That operator is handed a different incumbent, and on this instance exactly one incumbent, worth 107,195, ever leads it to 83,095, which is in turn the only route to the best solution we have recorded there. Every run reaching that solution had 83,095 in its log and every run that did not, did not. With the bound off we reproduce it three times in four; with it on, never in two.

The lesson generalises beyond this bound. Making a time-budgeted portfolio faster does not make it search longer along the same path, it makes it take a different path, and the new one need not be better. We priced that per instance, as the competition scores, and shipped with the bound disabled: -7.7% against $+2.2\%$ is a losing trade.

A better incumbent can repack worse. Incumbents worth 97,570, 100,685 and 107,195 yielded repacking improvements of 9,580, 13,510 and 24,100. A tight solution has no slack left, and this operator eats slack. Its result follows the incumbent's layout, not its objective. The roster nevertheless hands it the best-scoring solution every time.

Three changes that did not pay. The headroom from the faster conflict-graph build bought nothing: widening the 40-outsider candidate cap became affordable (331s of a 180s budget before, 179s after) and scored 3 better against 4 worse, median $+0.17\%$ — the evidence for wanting it had come from two instances. A free-column bitset for the packing search gave identical results and measured $1.6\text{--}1.9\times$ slower: its saving scales with columns per block, its cost with edges, and we had assumed the wrong line was hot. We also twice predicted the contact bound's benefit from its firing rate and were wrong in the same direction both times — most cells are rejected by a cheap bitmap probe, while the cells a bound removes are exactly those that would have reached the exact geometric test. Cell counts are not seconds.

2.4 Further Improvement Plan

The first item is to remove the wall clock from the schedule's decisions. We have localised where it enters. With the deadline lifted, the construction returns identical placements with and without the bound; under the deadline it receives, the two diverge. The divergence therefore begins in the construction's stopping

rule, not in the operator loop. The packing solver already takes an iteration cap for this reason. Once the schedule depends on work done instead of seconds elapsed, the contact bound can be re-enabled and contribute its speed rather than a trajectory.

Second, the repacking operator should sample the solution pool instead of always taking its best member. Its yield depends on slack, not on quality. The re-growth operator already samples the pool, so the mechanism is there to reuse.

Third, that operator’s cost model deserves the scrutiny its search has had. It declines a configuration whose predicted build will not fit the budget, and the prediction charges a fixed parsing cost to a quadratic term and mixes two different column counts. Both errors are squared, and both push towards declining work the budget could afford.

3 Implementation Details

Python 3.12 orchestrates and two C++17 extensions built with `pybind11` and OpenMP hold every hot loop. CP-SAT serves one operator and the algorithm degrades to the remaining five if it is unavailable. Python holds only control flow — the scheduler, the operator definitions, the acceptance test — while every geometric predicate, every candidate scan and both search engines are C++, the construction parallelising over beam states and the portfolio over worker processes.

The repacking operator. This operator lifts every block out of the most contested bay, adds the outside blocks that might want to enter it, and hands the set to a maximum-weight set-packing solver over space-time columns weighted in objective units. Destroying a bay and rebuilding it is ruin and recreate [4] with an exact recreate step, and the selection is a maximum-weight independent set searched by the swap-based scheme of Andrade et al. [1]. A call has two phases of different character: an uninterruptible build and a deadline-honouring search. The solver takes a total-time bound and enforces it against its own measured build cost rather than a predicted one; an earlier version subtracted a predicted build from the search budget as well, double-charging it. The configuration ladder is expressed as fractions of each instance’s own entry-time ceiling, so window widths come from the instance rather than being fitted to it.

Verification instrumentation. The engines carry counters for cells visited, exact tests, cheap rejections and cells skipped by the bound, plus a soundness mode that scores every cell the bound wanted to skip and counts those that would have won: zero out of 45.9, 11.6 and 9.3 million. An earlier bound had passed an identity check that was meaningless because the branch never executed, so we stopped accepting outcome comparisons as evidence.

Table 1. Preliminary-round training set, all forty instances at 60s. Sizes and bay counts are in Table 2.

#	Z_1	Z_2	Z_3	Obj.	#	Z_1	Z_2	Z_3	Obj.	#	Z_1	Z_2	Z_3	Obj.
1	0	157	2	1,499	15	0	4,366	155	42,445	29	0	6,364	1,212	382,692
2	0	144	15	3,690	16	0	3,769	145	38,130	30	123	573	1,442	1,930,651
3	0	2,281	137	43,360	17	0	6,575	213	54,629	31	233	2,807	10,151	5,825,327
4	0	2,278	0	15,946	18	0	4,098	218	45,386	32	411	1,925	2,337	2,781,688
5	0	1,834	228	47,038	19	0	4,350	321	60,093	33	839	1,666	4,591	6,298,923
6	0	2,746	133	39,172	20	0	3,693	618	99,408	34	102	1,970	1,368	1,082,900
7	0	2,514	254	55,698	21	19	1,530	2,202	598,927	35	26	3,991	1,547	598,663
8	0	2,413	8	11,252	22	1	2,955	1,792	738,998	36	28	4,383	6,813	111,628
9	0	2,147	265	50,485	23	90	93	1,560	1,532,961	37	585	1,783	3,579	4,104,337
10	0	1,419	450	69,783	24	2	4,235	717	262,941	38	2,306	3,683	5,788	32,489,664
11	0	305	137	20,356	25	266	2,772	1,662	213,434	39	391	1,219	5,062	5,977,379
12	0	2,385	450	76,545	26	440	1,216	5,100	6,640,032	40	2,216	3,156	8,947	1,597,539
13	0	3,327	476	79,943	27	1,442	22	4,182	20,899,030					
14	0	1,605	280	45,265	28	57	1,385	1,963	1,353,036					

4 Computational Results

Table 1 reports all forty preliminary-round training instances at a uniform 60s budget, decomposed into the three objective components; a uniform budget is the only comparable one, since the hidden limits vary and are not disclosed. All forty pass the provided feasibility checker, and the split is visible directly: Z_1 is zero on every instance from 1 to 20 and positive on the rest bar instance 29, where this build also reaches zero. Table 2 puts the two rounds side by side.

Across submissions. Table 3 gives what each of our three submissions returned on the eight final-round instances. It does not separate the changes from the noise. Every per-instance move is smaller than the spread the algorithm produces against itself; three replicates of one unchanged build spread 29.3%, against a largest move here of 7.2%. Both shipping steps also bundled two changes. What the table does record is direction. Five of eight instances improved from the first submission to the third, and the total is flat.

Per-seat pricing, on the final practice set. The change described in Section 2.2 was judged on all forty final practice instances, paired, at 180s, against two controls: no late window offered at all, and the late window offered under the old per-block weight. Judging by sign count over the set rather than instance by instance is deliberate. A single instance answers the same question with a spread of about 3%, and unchanged code shifted 8.4% between two runs an hour apart on this machine, so no single pair can resolve an effect of this size.

On the 35 instances whose weights permit a late window at all, the previous per-block weight scored 9 better against 21 worse (mean +0.62%) and the per-seat price 20 against 9 (mean -0.74%); on the four where no window can open

Table 2. The two training sets compared, both at 60s on the same build. They are different instances: w_1 spans 333 to 13,559 on the final set against 667 to 29,630 on the preliminary one, and w_3 reaches 800 against 600.

	Preliminary		Final	
	median	range	median	range
Blocks n	200	100–300	225	150–300
Bays m	3	2–5	3	2–5
Tardiness Z_1	0	0–2,306	480	1–10,536
Imbalance Z_2	2,333	22–6,575	2,425	91–15,458
Preference Z_3	668	0–10,151	4,210	196–10,614
Objective	105,518	1,499–32,489,664	5,005,623	6,804–75,792,403
Instances with $Z_1 = 0$	21 of 40		0 of 40	
Feasible	40 of 40		40 of 40	

Table 3. Objective returned on the eight final-round instances by each of our three submissions. Between the first and the second we shipped per-seat window pricing and the parallel conflict-graph build; between the second and the third, the beam salvage of Section 2.2 and the split-aim worker portfolio.

Instance	Submission 1	Submission 2	Submission 3	1 to 3
1	3,068,862	3,328,237	2,847,060	−7.2%
2	19,829,534	20,337,489	20,063,787	+1.2%
3	5,886,589	5,867,652	5,613,271	−4.6%
4	4,421,375	4,283,430	4,681,440	+5.9%
5	6,308,099	6,104,015	6,148,480	−2.5%
6	1,024,046	1,053,770	968,674	−5.4%
7	16,385,030	16,285,457	16,310,765	−0.5%
8	17,079,881	16,023,014	17,347,930	+1.6%
Total	74,003,416	73,283,064	73,981,407	−0.0%

it is 2–1 and does no harm, which was the condition we had set in advance for enabling it by default. The effect shrank as the sample grew, from −1.61% at twelve instances to −0.74% at thirty-nine: the first twelve had been chosen as the most favourable.

Two cautions belong with these numbers rather than after them. Run-to-run spread on the densest instance is about 112,000 on unchanged code, so smaller differences are noise. The sparsest is not stable at all: its objective is a minimum over the repacking operator’s attempts within a run, that operator’s improvement varies from 6,590 to 31,315 with the incumbent it is handed, and the value we quote appeared in four runs of seven.

Speed, measured at identical work. A fixed-time A/B cannot evaluate an accelerated search: both arms consume the same budget, so the faster one legitimately reaches a different solution. We replay one construction call with identical argu-

ments and no deadline, comparing a digest over every returned placement. On a sparse, a mid-density and a dense instance the bound skipped 40.8%, 3.4% and 1.3% of cells, none unsound, for $1.85\times$, $1.66\times$ and $1.09\times$ at an identical digest.

The conflict-graph build was accepted the same way, on an edge count confirmed unchanged on every configuration tested, for $10.5\times$ in isolation.

5 Team Members’ Contributions

The rules require anonymity, so members are identified by role. This section states who owned which decisions; Section 6 states how they were carried out. **Member A** owned the problem analysis and the algorithmic direction: treating geometry as a constraint on the objective rather than a term in it, choosing a measured-payoff portfolio over instance classification, setting what the contact bound had to satisfy before it could ship, and the reading of Section 2.3. **Member B** owned the codebase and the experimental discipline: version management, packaging, the rule that every comparison runs on an unloaded machine, which measurements were trustworthy enough to act on, and the check that every reported number matches the log behind it. Every change that shipped, and every change that did not, was decided by the authors.

6 AI Use Declaration

One generative AI tool was used: Anthropic Claude, through its command-line coding interface, as a coding and analysis assistant throughout development. The authors set the direction and the tool did the construction under it. Working from specifications settled in discussion, it implemented the two C++ extensions and the Python scheduler, worked out the algebra of the contact bound and the conditions under which it holds, built the measurement harnesses, ran the sweeps, diagnosed defects among them the position-dependent geometric predicate, and drafted this report for the authors to revise.

Nothing was adopted on the tool’s word. Every change was accepted or rejected by the authors against a measurement specified in advance, and several were rejected. The numbers here come from executing the submitted code, not from a language model.

References

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